REQUIREMENTS DOCUMENT

Pharmaceutical Field Force Management Platform

Mobile Application (Field) + Web Admin Console + Backend Platform

Functional & Non-Functional Requirements, Competitive Analysis, System Architecture and User Flows

Prepared: 19 August 2026

Prepared for: Konvert / Almashreq Soft — Internal Product & Client Pitch Use

Classification: Internal / Confidential — Draft v1.0

Document Control

Field	Detail
Document Title	Pharmaceutical Field Force Management Platform — Requirements Document
Version	1.0 — Draft
Status	For internal review; not yet baselined with a named client
Scope	Mobile field application, web admin console, and backend platform
Prepared Date	19 August 2026
Primary Market	Pakistan (pharmaceutical manufacturers and distribution enterprises)
Document Type	Business Requirements Document (BRD) + Software Requirements Specification (SRS) hybrid

How to use this document

This document is written to serve two purposes at once: (1) an internal requirements baseline the engineering and design team can build against, and (2) supporting material for client-facing pitches, since it also documents the competitive landscape and the reasoning behind key decisions. Section numbers are used as stable requirement references (e.g. “TP-03”) so they can be cited later in design documents, test cases and a formal traceability matrix without needing to renumber.

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1. Executive Summary

This document specifies the requirements for a pharmaceutical field force management platform: a mobile application used by Medical/Sales Representatives (MRs) in the field, a web admin console used by head office and management, and the backend platform connecting them. The goal is a system that lets a pharma company configure its own organisational hierarchy (organogram), plan and approve field representatives' monthly and daily work (tour planning), capture every doctor and chemist visit with GPS-verified accuracy (Daily Call Reporting), manage samples, orders and expenses, and give every level of management — from Area Manager up to National Sales Manager — real-time, hierarchy-scoped visibility into what is actually happening on the ground.

The requirements below are grounded in two things: (a) how the established category leaders — Veeva CRM, IQVIA's Orchestrated Customer Engagement (OCE) and Salesforce Life Sciences Cloud — structure this problem for large multinational pharma, and (b) how mid-market and Pakistan-specific competitors (RxRep, InstaCare, SyberEdge and regional South Asian SFA vendors) have adapted the same problem for the budget, connectivity and compliance realities of this market. Section 6 sets out that analysis in detail, and its central conclusion drives the scoping recommendation in Section 16: the enterprise-tier feature set is the wrong target for an MVP, and the realistic, winnable bar is to match the regional mid-market tier's functional depth while beating every current Pakistani competitor on GPS integrity, offline reliability, and configurability.

The document is organised so that Sections 2–4 establish context (purpose, methodology, personas, organogram); Section 6 covers the competitive landscape; Sections 8–9 are the requirements themselves (functional, then non-functional); Sections 10–11 describe the system architecture and data model needed to satisfy those requirements; Section 12 walks through the most important flows end-to-end (tour plan setup, a doctor visit, order booking, expense claims); and Sections 13–17 cover reporting, compliance, success metrics, a phased roadmap and concrete next steps.

2. Purpose, Scope & Objectives

2.1 Purpose

To define, in enough detail for design and engineering to estimate and build against, what the platform must do (functional requirements) and how well it must do it (non-functional requirements), and to document the market context that justifies those choices.

2.2 In Scope

- A native/cross-platform mobile app for field reps (Android primary, iOS supported).
- A web-based admin console for HQ, HR, Finance and management.
- A backend platform (API, database, sync engine, integrations) serving both front ends.
- All modules listed in Section 4 (organogram/user management through integrations).

2.3 Out of Scope (for this version of the document)

- Detailed UI/visual design (wireframes/mockups) — to follow as a separate design deliverable once this scope is signed off.
- Detailed pricing/licensing model for a SaaS offering — a commercial, not product-requirements, decision.
- Clinical trial, pharmacovigilance and regulatory-submission systems — this platform is commercial/field-force, not R&D or drug-safety tooling.

2.4 Objectives

- Give field reps a fast, mostly offline-capable tool so a full day's work (planning, visits, orders, expenses) can be logged in the field, not reconstructed from memory at home in the evening.
- Give every management tier real-time, hierarchy-scoped visibility into tour-plan adherence, doctor coverage, sales and expenses.
- Make the system configurable enough (hierarchy depth, visit norms, approval chains, expense slabs) to onboard different pharma clients without a code change per client.

- Build trust in the underlying field data through GPS integrity controls and full audit trails, since every downstream report and incentive payout depends on that data being real.

3. How This Document Was Prepared — Requirements-Gathering Methodology

This section is included deliberately: understanding how a requirements document like this is normally produced in the industry matters as much as the requirements themselves, both for judging how much confidence to place in this draft and for planning the next round of work with a real client.

3.1 The standard approach, and what this document does and does not replace

A production-grade BRD/SRS for a system like this is normally built through the following stages, and this document currently covers stage 1 (desk research) in full and stages 2–4 only partially (through competitor product literature and general industry benchmarks, not direct client interviews or field observation):

- Desk research & competitive analysis — what this document does in Section 6: study how established products (Veeva, IQVIA OCE, Salesforce Life Sciences Cloud) and direct competitors (RxRep, InstaCare, SyberEdge) solve the same problem, to avoid re-deriving well-established patterns from first principles.
- Stakeholder discovery interviews — structured 1:1 or small-group interviews with each user type: an NSM/Sales Director (strategic priorities, what reports they actually look at), a couple of RSMs/ABMs (approval pain points, how they currently monitor the field), several working MRs (daily friction, what takes too long today), and Finance/HR (expense policy, payroll integration needs, compliance obligations).
- Field shadowing ("day in the life" studies) — physically accompanying two or three MRs of different seniority/territory type for a full working day each. This is standard practice among sales-force-effectiveness consultancies precisely because a rep's actual day rarely matches the process described in a policy document — it surfaces the workarounds, dead zones (no signal), and shortcuts that a requirements doc built purely from interviews would miss.
- Requirements workshops (JAD sessions) — bringing the above stakeholders into a facilitated session to work through disagreements directly (e.g. Finance wants tighter expense controls; MRs want fewer approval steps) and reach an agreed priority, rather than the requirements team arbitrating alone.
- Documentation & prioritisation — writing the requirements down (as this document does) and prioritising with a method such as MoSCoW (Must/Should/Could/Won't — used throughout Section 8) or weighted scoring against business value and implementation cost.
- Traceability & sign-off — building a Requirement Traceability Matrix (RTM) linking each requirement to its source (which stakeholder/interview raised it), the design element that implements it, and the test case that verifies it, then walking every stakeholder through a formal sign-off before development starts. Section 17 recommends this as an immediate next step. See Section 3.3 for a starter RTM template.

3.2 Why this matters for this document specifically

Because stages 2–4 above have not yet been run against a specific client, the requirements in Sections 4 and 9 should be read as an evidence-based first draft, not a signed-off specification. They are grounded in what proven products and direct competitors already do (a defensible starting point), but a real client's organogram, approval culture, expense policy and existing systems will still change specifics — tier names, geofence radius, approval chain length, which integrations are actually needed on day one. Section 17 turns this into a concrete short list of next actions.

3.3 Starter Requirement Traceability Matrix (RTM) template

A traceability matrix is the standard mechanism for proving, at sign-off, that every requirement is actually going to be built and tested, and for tracing a bug or a change request back to its original source. A blank template, ready to populate once stakeholder interviews begin:

Req ID	Requirement Summary	Source (stakeholder/session)	Priority	Design Ref.	Test Case ID	Status
TP-04	Manager approval workflow for Monthly Tour Plan	(to be filled — e.g. ABM interview, 12 Aug)	Must	—	—	Draft

Req ID	Requirement Summary	Source (stakeholder/session)	Priority	Design Ref.	Test Case ID	Status
DCR-01	GPS geofenced check-in	(to be filled)	Must	—	—	Draft
EXP-02	Multi-level expense approval chain	(to be filled)	Must	—	—	Draft

4. Target Users & Personas

Five recurring personas appear across every competitor studied in Section 6, and the module list in Section 8 is organised so each persona's daily and weekly needs are covered end-to-end.

Persona	Primary Goal	Primary Device	Core Modules Used
Medical/Sales Representative (MR)	Cover the assigned doctor/chemist list efficiently, log visits fast, get expenses reimbursed without delay.	Android phone, often mid-range, frequently offline	Tour Plan, DCR, Samples, Orders, Expenses, Communication
Area Business Manager (ABM)	Keep the team's tour-plan adherence and DCR quality high; approve claims quickly without becoming a bottleneck.	Android phone / tablet	Approvals (Tour Plan, DCR exceptions, Expenses), Team Dashboard, Master-data approval
Regional / Zonal Sales Manager (RSM/ZSM)	Track territory-level target achievement and coverage; spot underperforming areas early.	Tablet / web	Roll-up Dashboards, Target Management, Territory Reports
National Sales Manager (NSM) / Director	National view of sales, coverage and cost; strategic territory and incentive decisions.	Web / desktop	Executive Dashboards, Incentive & Target configuration, BI export
Admin / IT / Compliance (Head Office)	Configure the system correctly for the organisation; keep master data clean; support audits.	Web / desktop	User & Org Management, Master Data, Business Rule Configuration, Audit Logs

5. Organisational Hierarchy (Organogram)

Pharma commercial organisations in Pakistan (and the wider region) overwhelmingly use a common escalating structure, even though exact titles vary by company. The platform must model this as configurable data, never as hard-coded tiers, since Section 4.1 (UM-01) exists precisely because every client's real organogram differs slightly.

Tier	Typical Title(s)	Typically Manages	Primary System Concern
1	Medical/Sales Representative (MR/SR)	A named list of doctors, chemists and stockists in one territory	Execute tour plan; log DCRs, samples, orders, expenses
2	Area Business/Sales Manager (ABM/ASM)	3–10 MRs across an area	Approve tour plans and exceptions; monitor DCR/expense compliance
3	Regional Sales Manager (RSM)	Multiple ABMs across a region	Regional target achievement; territory design decisions
4	Zonal Sales Manager (ZSM)	Multiple regions across a zone (a country is typically split into 4–6 zones)	Zonal strategy; cross-region resource allocation
5	National Sales Manager (NSM) / Sales Director	All zones	National sales strategy, incentive scheme design, board reporting

A visual organogram diagram (tree view of this structure, and how it maps to the system's role-based access model) is provided separately alongside this document.

6. Competitive Landscape Analysis

Rather than analysing a single competitor, this section deliberately spans three tiers: the global enterprise category leaders that set the industry's feature vocabulary, the regional South Asian mid-market vendors that are the realistic functional benchmark for an MVP, and the direct Pakistan-market competitors Konvert will actually be pitched against. Each is described in enough detail to be actionable, and Section 6.4 consolidates all of it into a single comparison table.

6.1 Global Enterprise Tier

Veeva CRM / Vault CRM

The long-standing category leader for large multinational pharma and biotech. It provides territory alignment (Veeva Align), compliant content delivery, sample accountability with full audit trails, and multichannel HCP engagement tracking, built specifically for the life-sciences regulatory environment. Veeva is now migrating its own customer base off Salesforce onto its proprietary Vault CRM platform, with Salesforce-based Veeva CRM entering a stability-only phase (security patches continuing, but no further feature development). This is relevant mainly as a signal of where the category is heading — tighter vertical integration between CRM, content and compliance — rather than as something Konvert needs to match feature-for-feature.

IQVIA Orchestrated Customer Engagement (OCE)

Positions itself as going beyond a traditional CRM by embedding real-time, AI/ML-driven intelligence directly into the rep's workflow — suggesting the next-best customer to target, the next-best content to send, and the next-best channel to use. It unifies sales, marketing and medical-affairs functions on a single Salesforce-based platform, and is used by over 350 life-sciences customers across 130+ countries. The “next-best-action” concept is worth tracking as a Phase-3 idea for Konvert (see CLM-04) once the core field-execution data is reliably flowing.

Salesforce Life Sciences Cloud

A newer, Salesforce-native alternative that a growing number of large biopharma organisations are choosing instead of migrating to Veeva Vault CRM, reflecting a genuine market split rather than one clear winner. It offers strong platform flexibility and a large systems-integrator ecosystem, but pharma-specific capability (field-force automation, sample accountability, consent management) is delivered through partner-built accelerators on top of the generic platform rather than being native out of the box — meaning real configuration effort and cost before it fits a pharma workflow.

Other named platforms worth being aware of

- Cegedim Relationship Management (Mobile Intelligence) — an earlier-generation life-sciences CRM whose technology was folded into IQVIA's OCE lineage; still referenced in the market as a legacy alternative.
- Aktana — an AI-layer vendor that sits alongside a CRM (rather than replacing it) to generate next-best-action suggestions for reps; a useful reference for what a future AI add-on to Konvert could look like without needing to be a full CRM replacement.

6.2 Regional (South Asian) Mid-Market SFA Tier

This tier — vendors such as SANeForce/MR Reporting, Bizzfield, TrackoField, Salestrip and EssentialSFA, all primarily India-based — is the most useful functional benchmark for Konvert's MVP, because it targets the same budget, connectivity and organisational scale as a typical Pakistani pharma company, rather than the enterprise scale Veeva and IQVIA are built for. Consistently, across every vendor in this tier, the same core capability set recurs: Monthly Tour Plan / Permanent Journey Plan (MTP/PJP) route planning, GPS-verified check-in/check-out for Daily Call Reports, offline-first data capture, expense/TA-DA claim automation, sample and stock tracking, and secondary-sales visibility down to the distributor level. Several vendors in this tier explicitly market DCR completion times under two minutes and describe eliminating the common failure mode of reps reconstructing the day's calls from memory hours later — both of which are strong, testable design targets Konvert should adopt directly (see DCR module and NFR-USE).

6.3 Direct Pakistan Market Competitors

Three Pakistan-based competitors surfaced directly during this research, and all three should be treated as active competitive references, not hypothetical ones.

RxRep (Lahore) — closest direct competitor

Marketed explicitly as “Pakistan's leading Field Force Management platform,” RxRep covers onboarding and hierarchy setup, territory definition, doctor and pharmacy assignment, role-based permissions, mobile DCR capture with GPS verification, prescription and sample recording, real-time sales and target-achievement dashboards, and an incentive-management module — essentially the same core scope this document defines in Section 4. Public marketing does not indicate a clear AI or advanced-analytics differentiator, which is a specific opening for Konvert given its stated AI-powered positioning.

InstaCare — adjacent Pakistan competitor

Primarily a healthcare/HR platform that has extended into pharma-specific HR and field-force tracking. Two details from its positioning are directly relevant to this requirements document and have already been reflected in Sections 4.11 and 9.10: it markets “WhatsApp-first” automated doctor follow-up (confirming WhatsApp is a real, expected local channel, not a nice-to-have), and it frames sample-distribution tracking explicitly around DRAP-compliant accountability reporting and FBR/EOBI/Labour-Law-compliant expense and payroll handling — useful confirmation of which local compliance hooks actually matter to buyers here.

SyberEdge — adjacent Pakistan competitor

A distribution-and-ERP-first platform for FMCG and pharma businesses, strong on van sales, FBR digital e-invoicing, GPS-tracked delivery, and Permanent Journey Plan (PJP)-based route/territory coverage. It competes more with Konvert's Order Management module (Section 4.7) and secondary-sales tracking than with the doctor-detailing side of the business, and is a useful reference for exactly what “FBR-compliant” needs to mean in practice (Section 9.10).

6.4 Comparison Matrix

Platform / Tier	Market Tier	Core Strength	Key Limitation for a Pakistani SME Client	Opportunity for Konvert
Veeva CRM / Vault CRM	Global enterprise	The long-standing category leader for large multinational pharma; deep HCP-engagement, compliant content delivery, sample accountability and territory-alignment tooling (Veeva Align). Now migrating off Salesforce onto Veeva's own Vault platform.	Priced and built for large multinational commercial organisations with big implementation teams — far beyond the budget, complexity tolerance and support model of a mid-size Pakistani distributor or manufacturer.	Konvert can offer 80% of the field-execution value (tour plan, DCR, GPS, territory, samples) at a fraction of the cost and implementation time, with local support and Urdu/English UI.
IQVIA Orchestrated Customer Engagement (OCE)	Global enterprise	Unifies sales, marketing and medical functions with AI/ML-driven “next-best-action” suggestions and strong omnichannel orchestration; built on Salesforce.	Same scale/cost mismatch as Veeva; also requires the client to already run (or license) IQVIA's broader data ecosystem to get full value.	Konvert should watch OCE's “next-best-action” direction as a Phase-3 roadmap idea, without chasing the full omnichannel/AI scope in an MVP.
Salesforce Life Sciences Cloud	Global enterprise	A newer, Salesforce-native alternative many large biopharma firms are now choosing instead of Veeva; strong platform flexibility and a large systems-integrator ecosystem.	Still a generic platform underneath — pharma-specific accelerators (field-force automation, sample accountability) are add-ons built by SIs, meaning real implementation cost and time.	Reinforces that a purpose-built, pharma-native product (like Konvert) removes the configuration burden a generic CRM leaves on the client.
Regional South-Asian SFA vendors (SANeForce/MR Reporting, Bizzfield,	Regional mid-market	Cover the same core ground Konvert needs — DCR, PJP/MTP tour plans, GPS visit verification, expense claims, sample	Built primarily for the Indian market: tax rules, distributor structures and support are India-centric, and none currently has a	Match this tier's feature depth (it is the realistic functional bar for an MVP) while winning on Pakistan-

Platform / Tier	Market Tier	Core Strength	Key Limitation for a Pakistani SME Client	Opportunity for Konvert
TrackoField, Salestrip, EssentialSFA and similar India-based tools)		tracking, secondary sales — at SME-friendly pricing, and are the closest functional benchmark for an MVP.	native Pakistani presence, DRAP/FBR alignment, or Urdu support.	specific compliance, local support and pricing in PKR.
RxRep (Lahore, Pakistan)	Direct Pakistan competitor	A Lahore-based platform marketed explicitly as "Pakistan's leading Field Force Management platform," covering user/hierarchy management, territory mapping, doctor and pharmacy databases, DCR with GPS, tour planning, sample and order management, and an incentive module.	Public marketing suggests a broad but fairly standard feature set with no visible AI/analytics differentiation; module depth and mobile app quality would need direct evaluation via a live demo.	Konvert's closest same-city competitor. Request a live demo/trial to benchmark UX and module depth directly, and differentiate on Konvert's stated AI angle plus deeper multi-tenant/incentive configurability.
InstaCare (Pakistan)	Adjacent Pakistan competitor	Primarily a healthcare/HR platform that has extended into pharma-specific HR and field-force tracking, explicitly marketed around DRAP-aligned sample accountability, FBR/EOBI-compliant payroll, and WhatsApp-first doctor follow-up.	Positioned more as HR/payroll-plus-tracking than a full commercial SFA (tour planning, order/secondary sales and incentive depth are not its core focus).	The WhatsApp-first communication pattern and DRAP/FBR compliance framing are worth adopting directly — they reflect genuine local market expectations, not just marketing language.
SyberEdge (Pakistan)	Adjacent Pakistan competitor	A distribution/ERP platform for FMCG and pharma with van-sales, FBR e-invoicing, GPS tracking and PJP-based route management — strong on the distributor/secondary-sales side.	Built distribution-first, not detailing/DCR-first — doctor engagement, sample accountability and medical-rep-specific workflows are not its focus.	A useful reference for the order-management and FBR e-invoicing requirements (Section 4.7/9.10), less so for the doctor-facing modules that are Konvert's core.

6.5 Key Takeaways & Positioning

- The enterprise tier (Veeva, IQVIA, Salesforce Life Sciences Cloud) is not the right competitive target for an MVP — the cost, implementation complexity and support model are built for multinational scale, not a mid-size Pakistani manufacturer or distributor.
- The regional South Asian mid-market tier is the correct functional bar: Konvert's MVP (Section 16) is scoped to match this tier's core capability, since that is what a realistic buyer will already be evaluating.
- RxRep is the most direct, same-market competitor and should be evaluated hands-on (request a live demo) before finalising Konvert's MVP scope, so gaps and differentiators are identified from firsthand product use, not marketing copy alone.
- DRAP-aligned sample accountability, FBR e-invoicing hooks, and WhatsApp-first communication are not optional local flourishes — every Pakistan-based competitor studied treats at least one of them as a headline feature, confirming they are genuine buyer expectations.
- No competitor studied — global or local — clearly leads on a modern, layered anti-GPS-spoofing defence (Section 9.5) despite it being a well-documented industry problem; this is a credible, technically achievable differentiator for Konvert rather than a marketing claim alone.

7. Assumptions & Constraints

7.1 Assumptions

- The first client(s) are pharmaceutical manufacturers or distribution enterprises operating in Pakistan with an existing MR/ABM/RSM-style hierarchy already in place informally (on paper or in Excel), not a brand-new sales organisation being designed from scratch.
- Field reps predominantly use company-issued or BYOD Android phones; iOS is a secondary requirement mainly for management-tier users.
- Network connectivity in the field is inconsistent, particularly outside major cities, making offline-first a hard requirement rather than a nice-to-have.
- A pilot client will be willing to participate in the stakeholder interviews and field shadowing described in Section 3 before scope is finalised.

7.2 Constraints

- Budget and timeline for an MVP are assumed to be materially smaller than an enterprise-tier deployment (see Section 6.5), which directly shapes the phased roadmap in Section 16.
- Any integration with a client's existing HRMS/ERP/payroll system is dependent on that system exposing a usable API or export mechanism — to be confirmed per client during onboarding.
- DRAP and FBR compliance requirements referenced in Section 9.10 are based on how existing local competitors publicly describe their own compliance posture, not on a legal review commissioned for this document; a compliance review is recommended before go-live (Section 17).

8. Functional Requirements

Requirements are grouped into 17 modules, numbered 4.1–4.17 for stable cross-referencing. Each requirement carries a MoSCoW priority (Must / Should / Could) as introduced in Section 3.1, reflecting a first-pass prioritisation to be revisited once stakeholder interviews are complete.

4.1 User, Role & Organogram Management

Covers how the client company's real-world reporting structure is modelled in the system, how accounts are provisioned, and how visibility of data is restricted by hierarchy.

ID	Requirement	Priority
UM-01	The system shall support a fully configurable, multi-tier organisational hierarchy (e.g. Medical/Sales Representative → Area Business Manager → Regional Sales Manager → Zonal Sales Manager → National Sales Manager → Sales Director), with tier names and depth configurable per client rather than hard-coded.	Must
UM-02	Admin shall be able to create, edit and deactivate user accounts capturing: full name, employee ID, designation, reporting manager, division/brand, headquarters town, joining date, contact number, CNIC, and bound device(s).	Must
UM-03	The system shall enforce role-based access control (RBAC) with predefined roles (MR, ABM, RSM, ZSM, NSM, Admin, HR, Finance, Super Admin) and allow an admin to define custom roles with granular, module-level permissions.	Must
UM-04	A manager's dashboards and reports shall be automatically scoped to their own downline/territory only, enforced server-side (never trusted from the client app).	Must
UM-05	The system shall support more than one concurrent hierarchy per employee where a rep sells more than one brand/division with different reporting lines.	Should
UM-06	The system shall provide a transfer/promotion workflow that reassigns territory, doctor lists and open approvals to the new owner while keeping historical DCRs, orders and expenses attributed to the employee who was in that role at the time (effective-dated history).	Must
UM-07	The system shall provide an interactive org-chart visualisation for HR/Admin, filterable by division, region and role.	Should
UM-08	Employees shall have a self-service profile screen (view-only) with a change-request flow for personal detail updates that requires HR approval.	Must
UM-09	The system shall allow a manager to nominate a delegate/backup approver for a defined date range (e.g. while on leave) so approval chains are never blocked.	Could
UM-10	The system shall provide an off-boarding workflow that revokes device/API access immediately, reassigns the open territory/doctor list to a successor, and retains full audit history against the departed employee's record.	Must

4.2 Territory, Area & Beat Management

Covers how the country is divided into sellable/coverable geography and how that geography is assigned to the field force.

ID	Requirement	Priority
TER-01	The system shall support a configurable geographic hierarchy: Country → Zone → Region → Area/Territory → Headquarters town → Beat.	Must
TER-02	Admin shall be able to assign one or more reps to a territory and split, merge or reassign territories with the change history retained against historical reporting.	Must

ID	Requirement	Priority
TER-03	The system shall provide a map-based territory visualisation showing coverage boundaries and doctor/chemist density.	Should
TER-04	The system shall support defining a Beat as a named, repeatable group of doctors/chemists visited on a fixed day-of-week or cycle.	Must
TER-05	The system shall allow a proposed territory realignment to be simulated (old vs new coverage, workload balance) before it is rolled out to the field.	Should
TER-06	Each territory shall carry its own target, doctor list and applicable product list, inherited by whichever rep is currently assigned.	Must
TER-07	The system shall provide a coverage-gap view (e.g. a heat-map of high-potential doctors that are under-visited) to support territory and beat redesign.	Could

4.3 HCP, Chemist & Stockist Master Data (MDM)

The customer master — doctors, chemists/pharmacies and distributors/stockists — is the foundation every other module (tour plan, DCR, orders) is built on, so its accuracy and governance matter more than any single feature.

ID	Requirement	Priority
MDM-01	The system shall maintain a doctor/HCP master with: name, specialty, qualification, hospital/clinic affiliation, address, geo-coordinates, potential grading (e.g. A/B/C), prescribing-behaviour tags, birthday, visit-frequency norm and consent status.	Must
MDM-02	The system shall maintain a chemist/pharmacy master with: name, address, geo-coordinates, licence number, linked stockist/distributor and credit terms.	Must
MDM-03	The system shall maintain a stockist/distributor master with: name, territory, contact, credit limit, running balance and linked warehouse.	Must
MDM-04	The system shall run duplicate-detection (name plus geo-proximity fuzzy match) whenever a rep proposes a new HCP or chemist record.	Must
MDM-05	New master records proposed from the field shall route through an approval workflow (MR proposes → ABM or Admin approves) before becoming visible platform-wide, to keep the master database clean.	Must
MDM-06	Admin shall be able to bulk import/export master data via Excel/CSV with row-level validation and an error report.	Should
MDM-07	The platform shall be architected to allow future sync with a third-party HCP reference database if the client licenses one, without a schema redesign.	Could
MDM-08	Every change to a master record shall be captured in an audit trail (who, when, before/after values).	Must
MDM-09	The system shall capture an HCP's communication consent/opt-out status and respect it when queuing digital outreach (WhatsApp, SMS, email).	Should

4.4 Tour / Journey Planning

This is the planning layer that turns a rep's assigned territory into a concrete, approved day-by-day schedule — the reference plan against which every day's field execution is later measured. See Section 12.1 for the full step-by-step user flow.

ID	Requirement	Priority
TP-01	A rep shall be able to create a Monthly Tour Plan (MTP) selecting headquarters/stations, working days, off-days and leave days for the upcoming month, submitted ahead of a configurable cut-off (e.g. 3 working days before month end).	Must
TP-02	A rep shall be able to define a Permanent Journey Plan / Standard Tour Plan (PJP/STP): a repeating weekly pattern mapping each day-of-week to a fixed beat (set of doctors/chemists).	Must
TP-03	The system shall auto-generate each day's plan from the PJP but allow the rep to edit it before execution, subject to an edit-lock window (e.g. editable only until the previous evening).	Must
TP-04	The Monthly Tour Plan shall route through a manager approval workflow supporting Approve, Approve-with-comments, or Reject-with-mandatory-reason.	Must
TP-05	The system shall validate, before submission, that the plan satisfies the minimum monthly visit-frequency norm per doctor category (e.g. Category A ≥ 3 visits/month) and block submission with an inline explanation if it does not.	Must
TP-06	The system should suggest an efficient visiting sequence for a given day's doctor list based on map distance/travel time.	Should
TP-07	The system shall produce a Planned-vs-Actual tour adherence report at rep, territory and national roll-up level.	Must
TP-08	The system shall support scheduling a joint field visit (manager accompanying a rep) against a specific date and beat.	Should
TP-09	Tour plans shall automatically exclude approved leave days and the relevant regional/provincial public holiday calendar.	Must
TP-10	The system may suggest an optimised tour plan using doctor potential, days-since-last-visit and prescription trend as inputs (later-phase AI capability).	Could

4.5 Daily Call Reporting (DCR) & Field Visit Execution

The point-of-visit capture module — arguably the single most-used screen in the entire application, and the one industry benchmarks judge hardest on speed and reliability. See Section 12.2 for the full step-by-step user flow.

ID	Requirement	Priority
DCR-01	A rep shall be able to check in to a call, capturing GPS coordinates and timestamp, validated against the master location within a configurable geofence radius (e.g. 100–200m).	Must
DCR-02	A check-in outside the geofence shall not be silently blocked; it shall be flagged for manager review and require the rep to enter a reason (e.g. “met doctor at the hospital, not the clinic”).	Must
DCR-03	During a call the rep shall record: products detailed, key messages delivered, samples/literature given, doctor feedback/objection, and the next-call commitment.	Must
DCR-04	The rep shall check out to close a call, with call duration captured automatically.	Must
DCR-05	The system shall optionally capture a digital acknowledgement (signature or OTP) from the HCP for sample receipt, per client policy.	Should
DCR-06	DCR entry and submission shall work fully offline, queued in an encrypted local store, and shall sync automatically with retry logic when connectivity resumes.	Must
DCR-07	A rep may amend a same-day DCR up to a configurable cut-off (e.g. midnight); after the cut-off, corrections require manager approval via an amendment request, and the original entry is preserved in the audit trail rather than overwritten.	Must
DCR-08	Managers shall have a dashboard showing DCR completion %, exceptions (geofence misses, anomalies) and daily team summaries.	Must

ID	Requirement	Priority
DCR-09	The rep shall be able to attach a geo- and time-stamped photo at point of call (e.g. shelf/POSM check, visiting card).	Should
DCR-10	The system should support voice-to-text note capture to speed up DCR entry.	Could
DCR-11	The system shall flag anomalies automatically: impossible travel speed between consecutive calls, calls logged during approved leave, or a mock-location signal on the device (see NFR-GEO in Section 9.6).	Must

4.6 Sample & Promotional Material Management

Sample accountability is both a productivity metric and a regulatory/audit concern, so stock movement must be traceable end-to-end from HQ to a named doctor.

ID	Requirement	Priority
SPL-01	The system shall maintain a rep-wise sample/gift stock ledger (opening balance, issued by HQ, distributed in field, closing balance) refreshed each allocation cycle.	Must
SPL-02	Sample distribution recorded on a DCR shall automatically decrement the rep's stock ledger against the specific doctor and product.	Must
SPL-03	The system shall produce a reconciliation report comparing samples issued vs distributed vs remaining, flagging variances for audit.	Must
SPL-04	The system should track sample batch numbers and expiry dates to prevent distribution of expired stock.	Should
SPL-05	The system may maintain a doctor-wise cumulative sample history to help enforce company distribution-cap policy.	Could

4.7 Order Management & Secondary Sales

Distinguishes primary sales (company → distributor) from secondary sales (distributor → chemist/retailer), which is the standard pharma commercial model and a frequently requested analytics cut.

ID	Requirement	Priority
ORD-01	A rep shall be able to capture a chemist/pharmacy order (product, quantity, applicable scheme/discount) during a field visit against a live price list.	Must
ORD-02	An order shall route to the chemist's linked distributor/stockist with status tracking: Booked → Confirmed → Dispatched → Delivered → Invoiced.	Must
ORD-03	The system shall capture secondary sales (distributor-to-retailer sell-out), distinct from primary sales, whether sourced from a distributor portal upload or rep-reported field data.	Must
ORD-04	The system should provide distributor stock/inventory visibility (real-time or periodic upload) to surface stock-outs early.	Should
ORD-05	The system should support a returns/expiry-claim workflow between chemist, distributor and company.	Should
ORD-06	The system may integrate with the client's ERP (SAP, Oracle, or a local Pakistani ERP) for invoicing and stock synchronisation.	Could

4.8 Expense Management (TA/DA)

Field expense claims are high-volume and, if slow, are one of the most common causes of MR dissatisfaction with an SFA rollout — so the workflow needs to be both fast for the rep and controlled for Finance.

ID	Requirement	Priority
EXP-01	A rep shall be able to submit expense claims (fuel/mileage, daily allowance, lodging, conveyance) mapped against policy slabs defined per grade and city category.	Must
EXP-02	Claims shall route through a configurable multi-level approval chain (e.g. ABM → RSM → Finance) supporting approve, query-back and reject.	Must
EXP-03	The system should auto-suggest mileage/fuel amount using the GPS distance actually logged between that day's calls, with the rep able to override with a reason.	Should
EXP-04	The system shall require a receipt photo attachment for claim types where company policy mandates it.	Must
EXP-05	The system shall produce a monthly settlement export for Finance/payroll, with a full audit trail per claim.	Must
EXP-06	The system should raise policy-violation alerts automatically (claim exceeds slab, duplicate claim, weekend claim requiring justification).	Could

4.9 Target Setting & Incentive Management

Ties day-to-day field activity back to the commercial numbers that ultimately justify the field force's cost, and is usually the module senior management cares about most.

ID	Requirement	Priority
TGT-01	Admin/HQ shall be able to set sales and activity targets at brand × territory × rep level for a defined period (monthly/quarterly/annual), cascading top-down and rolling up bottom-up.	Must
TGT-02	The system shall provide real-time target-vs-achievement dashboards at every level of the hierarchy.	Must
TGT-03	The system should provide a configurable incentive/commission engine (slabs, product weightage, qualifying conditions) with a payout preview.	Should
TGT-04	The system should support time-boxed sales contests/campaigns with a live leaderboard.	Should
TGT-05	The system may provide a what-if simulator to model incentive payouts before a scheme is finalised.	Could

4.10 E-Detailing & Closed-Loop Marketing (CLM)

A later-phase differentiator once the core field-execution modules are proven — digitises the sales-aid conversation itself and feeds engagement data back to marketing/brand teams.

ID	Requirement	Priority
CLM-01	The system should provide a versioned digital content library (approved sales aids, videos, brochures) centrally pushed to reps' devices.	Should
CLM-02	The system should support an in-call e-detailing presentation mode that logs slide-level engagement (which slides shown, time spent) automatically into the DCR.	Should
CLM-03	The system may support a content approval workflow (medical/legal/regulatory review) before material is published to the field.	Could
CLM-04	The system may suggest a “next-best-content” or “next-best-action” per doctor based on interaction history (AI/ML, later phase).	Could

ID	Requirement	Priority
CLM-05	Downloaded/cached content shall be fully usable offline, matching the DCR module's offline-first behaviour.	Must

4.11 Communication & Collaboration

Covers HQ-to-field and peer communication, reflecting that WhatsApp is already the de-facto channel between reps and doctors/managers in the Pakistani market.

ID	Requirement	Priority
COM-01	HQ shall be able to broadcast announcements/circulars to selected hierarchy segments with read-receipt tracking.	Must
COM-02	The system should provide in-app messaging between rep and manager, or a WhatsApp Business API hand-off, reflecting the channel reps already use with doctors and each other.	Should
COM-03	The system may send push notifications for pending approvals, target alerts, doctor birthdays and scheme launches.	Could

4.12 Attendance & Leave Management

Field attendance is naturally derived from the same GPS check-in used for DCR, avoiding a second parallel data source.

ID	Requirement	Priority
ATT-01	The system shall record daily geo-verified attendance mark-in/out, which may be tied to the first and last DCR of the day or captured as a separate action.	Must
ATT-02	The system shall support leave application with multi-level approval and running leave-balance tracking.	Must
ATT-03	The system should maintain a holiday calendar configurable per region/province, since public holidays vary across Pakistan's provinces.	Should
ATT-04	The system may integrate with the client's existing HRMS/payroll for attendance-linked salary processing.	Could

4.13 Reporting, Analytics & Dashboards

The layer that turns raw field data into decisions — every module above ultimately feeds this one.

ID	Requirement	Priority
RPT-01	The system shall provide role-scoped dashboards (MR: own performance; ABM/RSM: team roll-up; ZSM/NSM: regional/national) with drill-down to the underlying record.	Must
RPT-02	The system shall provide a standard report library: DCR compliance, tour-plan adherence, doctor coverage frequency, sample reconciliation, sales-vs-target, expense summary, secondary-sales trend.	Must
RPT-03	The system should provide a custom report builder with filters, scheduled delivery and export to Excel/PDF/CSV.	Should
RPT-04	The system should support embedding or export into an external BI tool (e.g. Power BI) for executive analytics, matching common industry practice.	Should
RPT-05	The system may provide predictive analytics such as churn-risk doctors (declining visit/Rx trend) or territory white-space opportunity.	Could

4.14 Approval Workflow Engine

A cross-cutting capability rather than a single screen — tour plans, expenses, master-data creation and leave all reuse the same configurable approval-chain engine instead of each being hand-built separately.

ID	Requirement	Priority
WF-01	The system shall provide a generic, configurable approval-chain engine reusable across Tour Plan, DCR exceptions, Expense Claims, Master-data creation and Leave.	Must
WF-02	The system shall support auto-escalation rules (e.g. escalate to the next level if not actioned within a configurable number of hours) so field work is never blocked by an unresponsive approver.	Must
WF-03	The system should support delegate-approver assignment for periods when the primary approver is unavailable.	Should

4.15 Learning & Training

Supports onboarding new joiners quickly — a real operational need given typical MR attrition/turnover in the industry.

ID	Requirement	Priority
LMS-01	The system may provide assignable product-knowledge modules and quizzes with completion tracking and certification.	Could
LMS-02	The system may provide an onboarding checklist for new joiners covering system, product and SOP training.	Could

4.16 Admin / Back-Office & Master Configuration

The control-plane an implementation team uses to stand up a new client without needing a code release for every business rule.

ID	Requirement	Priority
ADM-01	The system shall provide a central web-based admin portal for all master configuration: org hierarchy, territories, products, price lists, policies.	Must
ADM-02	The system shall support multiple brands/divisions within one client tenant, each able to have its own field force, products and targets.	Must
ADM-03	The platform should be built multi-tenant from day one so it can be sold as SaaS to more than one pharmaceutical client company on shared infrastructure, with strict data isolation between tenants.	Should
ADM-04	Business rules (visit-frequency norms, geofence radius, expense slabs, approval chains, edit-lock windows) shall be configurable by an admin without a code deployment.	Must
ADM-05	The system shall provide data import/export and migration tooling to support onboarding a new client's existing master data.	Must

4.17 Integrations

Lists the external systems a real pharma client will expect the platform to talk to, prioritised by how commonly they appear in the Pakistani market context.

ID	Requirement	Priority
INT-01	The system should integrate with the client's HRMS/payroll for attendance, leave and expense settlement.	Should
INT-02	The system should integrate with the client's ERP/accounting system for order-to-invoice flows.	Should
INT-03	The system may integrate with the WhatsApp Business API for doctor communication and internal notices, reflecting an already-established local usage pattern.	Could
INT-04	The system may integrate with an SMS gateway for OTP and critical alerts in low-connectivity areas.	Could
INT-05	The system should integrate with a maps/GIS provider for territory mapping and route optimisation.	Should
INT-06	The system should provide hooks to support FBR-compliant e-invoicing where the platform touches the order-to-invoice flow.	Should
INT-07	The system may integrate with a BI tool (Power BI or similar) for embedded executive analytics.	Could

9. Non-Functional Requirements

Non-functional requirements are grouped into 13 categories. In a field-force system, three of these — offline architecture (9.4), location integrity (9.5) and usability (9.7) — tend to determine adoption success or failure more than any single functional feature, since a rep who cannot trust the app to work without signal, or who finds DCR entry too slow, will revert to paper or WhatsApp within weeks regardless of how complete the feature list is.

9.1 Performance

- Cold app start under 3 seconds on a mid-range Android device (~2GB RAM); screen transitions under 1 second.
- API 95th-percentile response time under 500ms under normal load; bulk/report operations may run asynchronously with a progress indicator.
- The backend shall support at least 500 concurrent active field users per tenant without perceptible degradation, scaling further via horizontal scaling as tenants and headcount grow.

9.2 Scalability

- A stateless API layer with a managed, horizontally scalable database so the platform can grow from a single pilot client (~50 users) to a multi-tenant SaaS platform serving many pharma companies and thousands of field reps nationally.
- A data partitioning/archival strategy for large HCP master data and multi-year DCR/order history so query performance does not degrade over time.

9.3 Availability & Reliability

- Target backend uptime of 99.5%+ in production, with scheduled maintenance windows communicated to clients in advance.
- The mobile app shall remain fully usable for core field actions (tour plan, DCR, orders, expenses) with zero connectivity for at least 5–7 working days, syncing automatically once connectivity resumes — essential given patchy network coverage in many smaller Pakistani towns and rural territories.

9.4 Offline-First Architecture & Sync

- An encrypted local on-device database mirroring the subset of master data relevant to that rep's territory.
- A defined conflict-resolution policy for offline edits that later conflict with server state (e.g. rep-authored records are rep-authoritative until submitted; master data is always server-authoritative).
- Delta-sync (only changed records) rather than a full re-download on every sync, to conserve mobile data on cost-sensitive plans.
- Background sync plus a manual “Sync now” control, with a visible per-record sync-status indicator so the rep always knows what has and has not reached the server.

9.5 Location Integrity (Anti-Spoofing)

- Multi-layer GPS spoofing/mock-location defence: device mock-location-flag detection, root/jailbreak and emulator detection, and physics-based anomaly checks (impossible speed between two points, unnaturally constant GPS accuracy) — the layered approach the industry now treats as standard practice.
- Server-side secondary validation of location data, since app-level checks alone can be bypassed on a rooted device.
- Flagged/suspicious location events shall be retained — never silently discarded — with a full audit trail for manager review, and shall be excluded from KPI/incentive calculations until cleared.

9.6 Device & Platform Compatibility

- Cross-platform mobile app (e.g. Flutter or React Native) for Android as the primary platform (minimum Android 8.0/Oreo) given Android's dominant field-force share locally, with iOS supported for management-tier users.
- The app must run acceptably on low-to-mid-range Android devices (2–3GB RAM) commonly issued to field staff, not only flagship phones.
- The admin web portal shall be responsive and support the latest two versions of Chrome, Edge, Firefox and Safari.

9.7 Usability

- A DCR entry shall be completable in under 2 minutes per call — a widely cited industry benchmark — through wizard-style flows, pre-filled defaults and minimal typing.
- Bilingual UI (English default, Urdu optional) to suit varying field-force comfort levels across regions.
- Outdoor-readable design (adequate contrast, large tap targets) since the app is used in direct sunlight far more often than a typical office app.

9.8 Localisation

- PKR currency formatting, Pakistani date conventions, and configurable multi-province public-holiday calendars.

9.9 Security

- Role-based access control enforced at both UI and API layer — the client app is never trusted as the sole gatekeeper.
- Data encrypted at rest (AES-256) and in transit (TLS 1.2+).
- Secure authentication (password plus OTP or device biometric), session timeout, and device-binding so a login cannot silently be used from an unrecognised device.
- Full audit logging (who, what, when, from where) on all sensitive entities: DCR, expenses, master data, user roles and approvals.
- API-layer protections against common threats: rate limiting, input validation and OWASP Top 10 mitigations.
- Certificate pinning and root/jailbreak warnings to reduce the risk of tampered location or sales data reaching the server.

9.10 Regulatory & Compliance

- Sample-distribution records maintained with the accountability and audit trail expected under DRAP's (Drug Regulatory Authority of Pakistan) oversight of therapeutic-goods distribution and promotion.
- FBR-compliant e-invoicing hooks wherever the platform touches order-to-invoice, consistent with how comparable local distribution platforms already operate.
- Data-handling practices that anticipate Pakistan's evolving personal-data-protection framework (consent capture, data-subject access), so the platform does not need re-architecture if/when a binding law takes effect.
- A configurable data-retention policy per client (commonly 3–7 years for DCR/expense audit trail).

9.11 Maintainability

- A modular, API-first codebase so the mobile app and any future channel (e.g. a self-service web portal) consume the same documented backend contracts.
- An automated regression suite covering the core flows (tour plan, DCR, sync, approvals) run before every release.
- A CI/CD pipeline with staged rollout: internal → pilot client → general availability, including phased app-store releases.

9.12 Backup & Disaster Recovery

- Automated daily backups with a Recovery Point Objective of ≤24 hours and a Recovery Time Objective of ≤4 hours.
- Geographically separated backup storage with periodic restore-drill testing.

9.13 Auditability

- An immutable audit trail for regulated actions — sample distribution, post-cutoff DCR amendments, master-data changes, approval decisions — retained per the client's data-retention policy and exportable for a regulatory or internal-audit inspection.

10. High-Level System Architecture

The architecture below is a recommended starting point that satisfies the non-functional requirements in Section 9 (particularly offline-first sync and multi-tenant scalability); it is intended to guide estimation and technical design, not to lock in a final technology choice.

10.1 Component overview

- Field Mobile App — cross-platform (Flutter or React Native recommended for a single codebase across Android/iOS) with a local encrypted database (e.g. SQLite/Realm/WatermelonDB) implementing the offline-first sync described in NFR 9.4.
- Web Admin Console — a responsive single-page application (e.g. React/Next.js) for HQ, HR, Finance and management users, consuming the same backend API as the mobile app.
- API Layer — a versioned REST or GraphQL API (e.g. Node.js/NestJS or a similarly productive framework) that is the single source of truth both front ends talk to; all RBAC and hierarchy-scoping (UM-04) is enforced here, never trusted from either client.
- Sync Engine — a dedicated delta-sync service handling conflict resolution (per NFR 9.4), queued offline writes, and background push of master-data updates to devices.
- Core Database — a relational database (e.g. PostgreSQL) for transactional data (users, territories, DCRs, orders, expenses), chosen for strong consistency guarantees on financial and audit-sensitive records.
- Analytics/Reporting Store — a read-optimised store or data warehouse (could be a replica, or a dedicated analytics database) feeding the dashboards in Section 4.13, kept separate from the transactional path so heavy reports never slow down field-app writes.
- Object Storage — for photos, receipts, e-detailing content and exported reports (e.g. S3-compatible storage).
- Location Integrity Service — a dedicated backend service implementing the server-side half of the anti-spoofing checks in NFR 9.5, independent of whatever the mobile app itself reports.
- Notification Service — push notifications, SMS gateway and WhatsApp Business API integration (INT-03, INT-04, COM-03).
- Integration Layer — adapter services for HRMS, ERP and BI-tool integrations (Section 4.17), isolated from the core so a client-specific integration never touches core platform code.

10.2 Multi-tenancy

Given ADM-03 (should support SaaS delivery to multiple pharma clients), the recommended approach is logical multi-tenancy within a shared database (a tenant_id on every table plus row-level security), rather than a separate database per client, to keep operational overhead manageable while strictly isolating each client's data at the query layer. This should be validated against the client base's compliance expectations — some clients may require or prefer physical data separation.

10.3 Data flow: a single field visit, end to end

- 1. Rep checks in on the mobile app → write held in local encrypted store immediately (works even with zero signal).
- 2. On reconnect, the Sync Engine pushes the queued DCR to the API layer.
- 3. The Location Integrity Service independently validates the GPS data server-side.
- 4. The API layer writes the validated record to the Core Database and raises an event.
- 5. The event updates the manager's live dashboard (Analytics Store) and, if the visit was flagged, adds it to the exception review queue.
- 6. Any sample distributed against that call decrements the rep's Sample Ledger (SPL-02) in the same transaction, keeping stock and activity data consistent.

11. Data Model / Key Entities

A conceptual entity overview, sufficient for architecture and API design; a full entity-relationship diagram with field-level detail should be produced during technical design.

Entity	Key Attributes	Key Relationships
User (Employee)	name, employee ID, role, designation, device(s), status	belongs to OrgUnit/Territory; self-referencing reports_to
OrgUnit / Territory	name, tier, geography, parent territory	has many Users; has many Beats
HCP (Doctor)	name, specialty, category, geo-coordinates, consent status	belongs to Territory; has many DCRs
Chemist / Pharmacy	name, address, geo-coordinates, licence no.	belongs to Territory; linked to a Distributor; has many Orders
Distributor / Stockist	name, credit limit, balance, warehouse	has many Chemists; has many Orders (secondary sales)
Product	name, brand/division, price, active status	belongs to Division; referenced by DCR, Order, Sample Ledger
Tour Plan (Monthly) → Day → Planned Visit	month, status, per-day station/beat	belongs to User; generates DCRs on execution
DCR (Call)	check-in/out GPS+time, products detailed, outcome	belongs to User; links HCP/Chemist; links Planned Visit (optional)
Sample Ledger Entry	product, quantity, batch, direction (issued/distributed)	belongs to User; linked to a DCR when distributed
Order	products, quantities, scheme, status	belongs to Chemist and Distributor; linked to the DCR it was booked on
Expense Claim	type, amount, date, receipt, status	belongs to User; routed through Approval Workflow
Target	period, brand, territory/user, value	belongs to Territory/User and Product
Approval Workflow Instance	entity type + ID, chain, current step, status	generic — reusable across Tour Plan, DCR exception, Expense, Master data, Leave
Audit Log Entry	entity, action, before/after, actor, timestamp, location	references any auditable entity above

12. Detailed User Flows

The four flows below are written in a standard use-case format (actor, preconditions, main flow, alternate/exception flow, postconditions) so they can be handed directly to design and QA. Flow 12.1 responds specifically to how a rep sets up and executes a tour plan, since this is the anchor workflow every other field module (DCR, orders, expenses) is scheduled around.

12.1 Flow: Setting Up and Executing a Monthly Tour Plan

Actor: Medical Representative (MR). Approver: Area Business Manager (ABM). Escalation: Regional Sales Manager (RSM).

Preconditions

- The MR is an active user assigned to a territory that already has a Permanent Journey Plan (PJP) — a repeating weekly beat pattern — configured, either by the MR previously or seeded by Admin during onboarding.

Main flow

- 1. Trigger: on a configurable date near month-end (e.g. the 3rd-to-last working day), the system sends a push notification and in-app banner prompting the MR to submit next month's Tour Plan.
- 2. The MR opens the Tour Plan module and selects the upcoming month. The system auto-populates a day-by-day calendar using the existing PJP, and automatically greys out Sundays, the region's public holidays, and any already-approved leave days for that MR.
- 3. For each remaining working day, the MR reviews the suggested station/beat and either: (a) accepts it as-is, (b) swaps in a different station or beat for that day (e.g. a planned outstation trip), or (c) marks the day as off/leave — which the system cross-checks against the MR's leave balance and flags if no leave application exists.
- 4. The MR may attach a special activity to a specific date — a CME event, a medical camp, or a joint field visit with the ABM — distinct from a normal beat day.
- 5. Before the plan can be submitted, the system validates: minimum monthly visit-frequency compliance is achievable per doctor category (TP-05), the total planned working days match the expected norm for that month, and no day has been left without a status. Any violation is shown inline with an explanation rather than a generic error, and blocks submission until resolved or explicitly justified.
- 6. The MR submits the plan. Status changes to “Pending Approval” and the ABM is notified.
- 7. The ABM opens their approval queue and reviews the plan alongside last month's actual coverage gaps and the doctor-visit-norm heat-map for that territory, plus any violation the MR justified rather than fixed.
- 8. The ABM can Approve as-is, Approve with inline comments/change requests against specific dates, or Reject with a mandatory reason, which returns the plan to the MR to revise (loops back to step 3).
- 9. On approval, the plan's status becomes “Approved” and it becomes the reference plan for the month. Each day, the system generates that day's “Today's Plan” screen in the MR's app, available from the previous evening or that morning.
- 10. Each morning, the MR's home screen shows an ordered list of that day's planned doctors/chemists at the relevant station. The MR may reorder the list or add an unplanned/ad-hoc call, which is captured with an “unplanned visit” flag for later variance reporting rather than silently blended into the plan.
- 11. As the day progresses, each completed call updates a live “planned vs completed” progress indicator for that day, visible to the MR and, in aggregate, to the ABM.
- 12. At day-end, the system computes that day's Tour Plan Adherence % (calls completed from the plan ÷ calls planned), rolling up into weekly and monthly adherence visible up the full hierarchy (ABM → RSM → ZSM → NSM) with drill-down to the underlying day.

Alternate / exception flow

- Mid-month change: if the MR needs to swap a day (e.g. a doctor becomes unavailable), the same lightweight approval loop (steps 6–8) applies to just that date.

- Approver non-response: if the ABM does not act within a configurable window (e.g. 48 hours), the system auto-escalates the pending plan to the RSM per the escalation rule in WF-02, so the MR is never blocked from starting the month.

Postconditions

An approved, dated Tour Plan exists as the reference against which the month's Daily Call Reports and adherence reporting are measured; every deviation from it is captured and auditable rather than silently absorbed.

12.2 Flow: A Doctor Visit → Daily Call Report

Main flow

- 1. The MR arrives at the doctor's clinic and taps “Start Call” against that day's planned entry, or “Ad-hoc Call” if it was not in the plan.
- 2. The app captures GPS and compares it to the doctor's stored location within the configured geofence radius. Inside the radius, check-in is accepted immediately. Outside it, the app prompts the MR to confirm the location or provide a reason (e.g. “met the doctor at the hospital, not the listed clinic”); the call proceeds either way but is flagged for manager review rather than blocked outright — keeping the app usable in the messy reality of field work while still preserving oversight.
- 3. The MR selects the products detailed from that division's active product list and tags the call's objective (e.g. new-Rx ask, follow-up, objection handling).
- 4. If samples or literature are given, the MR selects them from their current stock, which auto-decrements the Sample Ledger (SPL-02); a photo may be captured if company policy requires proof.
- 5. The MR records a short note or voice note on the doctor's feedback and the agreed next step.
- 6. The MR taps “End Call”; duration is logged automatically and the DCR is saved locally — this works with zero connectivity — and queued for sync.
- 7. Once connectivity is available, the DCR syncs to the server in the background; the MR sees a clear per-record status (queued / syncing / synced / failed — retrying), so nothing is silently lost.
- 8. The synced DCR appears on the ABM's live feed within minutes; any flagged exception (geofence miss, anomaly) lands in a dedicated review queue rather than being buried in the normal feed.
- 9. The MR may amend the DCR until the same-day cut-off. After that, a correction must go through an amendment request to the ABM, and the original entry is preserved in the audit trail rather than overwritten (DCR-07).

12.3 Flow: Chemist Order Booking → Secondary Sales

- 1. The MR visits a chemist and opens the order screen, which loads the live price list and any active trade scheme.
- 2. The MR adds products and quantities; applicable discounts/schemes are applied automatically rather than calculated by hand.
- 3. The order is submitted and routed to the chemist's linked distributor/stockist.
- 4. The distributor accepts, modifies or rejects the order (via a distributor portal or a manager relay where the distributor has no direct system access); status updates flow back to the MR and ABM.
- 5. On dispatch/invoicing — via ERP integration where connected — the order closes and contributes to secondary-sales analytics and the MR's booking-target achievement.

12.4 Flow: Expense Claim Submission

- 1. The MR opens the Expense module, adds a claim, and selects the date and type (fuel, daily allowance, lodging, conveyance).
- 2. The system pre-suggests a distance/fuel amount using that day's logged GPS trail between calls; the MR can adjust it, but must give a reason if the adjustment is significant.
- 3. A receipt photo is attached where the claim type requires one.
- 4. The claim is submitted and routed through the configured chain (typically ABM → RSM → Finance); each approver can approve, query back for clarification, or reject with a reason.

- 5. Approved claims batch into a monthly settlement export for Finance/payroll (EXP-05).

13. Reports & KPI Catalogue

The standard report library referenced in RPT-02, expanded with the specific KPI each report is built to answer.

Report	Primary KPI(s)	Primary Audience
Tour Plan Adherence	% of planned calls actually completed, by day/week/month	ABM, RSM, ZSM, NSM
DCR Compliance & Quality	% of days with a submitted DCR; % flagged as an exception	ABM, RSM
Doctor Coverage Frequency	Actual visits per doctor per month vs the category norm	ABM, RSM, ZSM
Sample Reconciliation	Samples issued vs distributed vs remaining, by rep	ABM, Compliance, Admin
Sales vs Target	Primary/secondary sales achievement % by rep, territory, product	All management tiers
Secondary Sales Trend	Distributor-to-retailer sell-out trend by territory/product	RSM, ZSM, NSM
Expense Summary	Claimed vs approved amount by type, rep, territory	Finance, RSM
GPS Exception Log	Geofence misses, anomalies, flagged mock-location events	ABM, Compliance
Rep Productivity Scorecard	Calls/day, coverage %, sales/call, expense/call composite view	ABM, RSM, HR

14. Regulatory & Compliance Considerations (Pakistan)

These are the local regulatory touch-points identified during the competitive research in Section 6 — every one of them is already treated as a headline feature by at least one Pakistan-based competitor, indicating they are real buyer requirements rather than optional extras. A formal legal/compliance review with counsel familiar with DRAP and FBR requirements is recommended before go-live (Section 17).

14.1 DRAP (Drug Regulatory Authority of Pakistan)

DRAP regulates the manufacture, distribution, promotion and sale of therapeutic goods in Pakistan. For this platform, the practical implication is that sample distribution and promotional-material usage need the kind of accountable, auditable trail already required in Section 4.6 (Sample & Promotional Material Management) and Section 9.13 (Auditability) — a complete, exportable record of what was distributed, to whom, by whom, and when.

14.2 FBR (Federal Board of Revenue)

Wherever the platform touches order-to-invoice (Section 4.7), FBR's digital e-invoicing requirements need to be supported, matching the pattern already implemented by SyberEdge in the same market (Section 6.3). Expense settlement exports (EXP-05) should also be structured to align with standard payroll/tax documentation practice.

14.3 Data Protection

Pakistan's personal-data-protection framework is still evolving. The recommended approach, consistent with NFR 9.10, is to build in consent capture and reasonable data-subject controls now, so the platform does not need re-architecture if and when a binding law takes effect — this is materially cheaper than retrofitting later.

14.4 Provincial Variation

Public holidays, and in some cases specific labour or tax rules, vary by province. The holiday-calendar configurability in TP-09 and ATT-03 exists specifically to accommodate this without a code change per region.

15. Success Metrics for the Platform Itself

Metrics to judge the platform's own success post-launch, distinct from the sales KPIs it reports on (Section 13).

Metric	Target	Why it matters
Daily active field-user rate	≥ 90% of licensed reps active on a working day	Core adoption signal — an unused app produces no data
Average DCR entry time	< 2 minutes per call	Matches the industry benchmark cited in Section 6.2; the single biggest driver of daily adoption
Same-day DCR submission rate	≥ 95%	Indicates reps are logging in real time, not reconstructing from memory
Sync failure rate	< 1% of records requiring manual retry	Direct measure of offline-sync reliability (NFR 9.4)
GPS exception rate	Tracked and trending down, not zero (zero is itself suspicious)	A healthy exception rate shows the anti-spoofing system (NFR 9.5) is actually catching something
Tour plan approval turnaround	< 24 hours median	Approval-chain bottlenecks are one of the most common reasons SFA rollouts stall
Expense claim settlement time	< 7 working days end-to-end	A frequently cited driver of field-force dissatisfaction with SFA tools generally

16. Implementation Roadmap

A phased approach, sequencing the modules in Section 8 so that the highest-adoption-risk capabilities (offline DCR, tour planning, GPS integrity) are proven early with a real pilot client, before layering on the commercially richer but lower-adoption-risk modules.

16.1 Phase 0 — Discovery (2–4 weeks)

- Run the stakeholder interviews and field-shadowing described in Section 3.2 with a named pilot client.
- Confirm the client's actual organogram, territory structure, and existing systems (HRMS/ERP) to integrate with.
- Run a hands-on evaluation of RxRep (Section 6.3) via a live demo to sharpen Konvert's differentiation before locking MVP scope.
- Finalise the MoSCoW priorities in Section 8 against the pilot client's actual priorities and populate the Requirement Traceability Matrix (Section 3.3).

16.2 Phase 1 — MVP: Core Field Execution

- Modules: User/Role/Organogram (4.1), Territory Management (4.2), HCP/Chemist Master Data (4.3), Tour Planning (4.4), Daily Call Reporting (4.5), basic Reporting (4.13, Must items), Approval Workflow Engine (4.14), core NFRs (offline sync, GPS integrity, security).
- Goal: prove reps will actually use the app daily, and that management trusts the data it produces, before adding commercial complexity.

16.3 Phase 2 — Commercial Depth

- Modules: Sample Management (4.6), Order Management & Secondary Sales (4.7), Expense Management (4.8), Target & Incentive Management (4.9), Attendance & Leave (4.12), expanded Reporting.

16.4 Phase 3 — Differentiation & Scale

- Modules: E-Detailing/CLM (4.10), advanced Communication (4.11), Learning & Training (4.15), full Integration layer (4.17), predictive/AI analytics (RPT-05, CLM-04, TP-10).
- This is also the natural point to formalise multi-tenant SaaS delivery (ADM-03) if Konvert intends to sell to multiple pharma clients on shared infrastructure.

17. Recommended Next Steps

- 1. Identify and confirm a pilot client, and schedule the stakeholder interviews and field-shadowing described in Section 3.2 — this is the single highest-value next action, since it will validate or adjust nearly every assumption in Sections 4–9.
- 2. Request a live demo or trial of RxRep specifically, and if feasible InstaCare, to benchmark UX, module depth and pricing hands-on rather than from marketing material alone.
- 3. Populate the Requirement Traceability Matrix (Section 3.3) as interviews happen, and run a formal prioritisation workshop (JAD session, Section 3.1) with the pilot client to re-confirm or adjust the MoSCoW priorities in Section 8.
- 4. Commission a short legal/compliance review specifically covering DRAP sample-accountability expectations and FBR e-invoicing requirements (Section 14), since both directly shape the data model (Sections 4.6, 4.7, 11).
- 5. Produce UI/UX wireframes for the Phase 1 MVP modules (Section 16.2), starting with the Tour Plan and DCR flows in Section 12, since these are the screens a rep will open most often and therefore the ones most worth getting right first.
- 6. Stand up the core architecture (Section 10) with the offline-sync engine and GPS-integrity service built and tested first, ahead of any UI work — these are the two components hardest to retrofit later and most responsible for whether the app is trusted in daily use.
- 7. Define the specific configurable defaults (geofence radius, DCR edit-lock time, visit-frequency norms, expense slabs, escalation windows) with the pilot client before development starts, since these numbers appear throughout Sections 4 and 9 as “configurable” but still need a sensible starting value.
- 8. Plan a pilot rollout with a small subset of the client's field force (e.g. one territory or ABM's team) before a full national rollout, so issues with adoption, connectivity or data quality are caught while the blast radius is still small.

18. Glossary of Terms

Term	Definition
MR / SR	Medical Representative / Sales Representative — the field-level rep who visits doctors and chemists.
ABM / ASM	Area Business/Sales Manager — first-line manager, typically overseeing a handful of MRs in a defined territory.
RSM / RM	Regional Sales Manager — manages multiple ABMs across a region.
ZSM	Zonal Sales Manager — manages multiple regions within a zone.
NSM	National Sales Manager — owns the field force nationally.
HCP	Healthcare Professional — typically a doctor, in this context.
DCR	Daily Call Report — the record of a single field visit/call.
PJP / STP	Permanent Journey Plan / Standard Tour Plan — a repeating weekly beat pattern.
MTP	Monthly Tour Plan — the approved, dated plan for a calendar month, derived from the PJP.
POB	Primary Order Booking — an order captured from a chemist/pharmacy at point of visit.
Rx	Prescription — shorthand for a doctor's prescribing activity/behaviour.
Primary sales	Sales from the pharma company to its distributor/stockist.
Secondary sales	Sales from the distributor/stockist onward to the chemist/retailer.
CLM	Closed-Loop Marketing — digital, trackable detailing content whose usage data feeds back to brand/marketing teams.
Geofencing	Validating a GPS check-in against an allowed radius around a stored location.
Ghost call	A DCR logged for a visit that did not actually happen at the stated location/time.
MoSCoW	A prioritisation method: Must have, Should have, Could have, Won't have (this release).
DRAP	Drug Regulatory Authority of Pakistan — national regulator for therapeutic goods.
FBR	Federal Board of Revenue — Pakistan's tax authority, relevant to e-invoicing requirements.

19. References

Sources consulted during the competitive and industry research underlying this document. All product descriptions above are paraphrased summaries of publicly available material, not reproductions.

- Veeva CRM / Vault CRM — veeva.com (product pages, field-force-planning and territory-alignment blog posts)
- IQVIA Orchestrated Customer Engagement (OCE) — iqvia.com (product fact sheet and case studies)
- Salesforce Life Sciences Cloud vs Veeva — customertimes.com; apptad.com (industry transition analysis)
- RxRep — rxrep.net (Pakistan field force management platform)
- InstaCare — instacare.com.pk (Pakistan pharma HR/CRM platform)
- SyberEdge — syberedge.com (Pakistan distribution/ERP platform)
- Regional SFA vendors — saneforce.com / mrreporting.saneforce.com, bizzfield.com, trackobit.com, salestripsfa.com, essentialsfa.com, salesjump.in
- Closed-loop marketing / e-detailing — avenga.com, viseven.com, enablrsales.com
- GPS spoofing / mock-location detection — approov.io, guardsquare.com, deepidsdk.com, wappblaster.com
- Pharma sales hierarchy (MR/ABM/RSM/ZSM/NSM) — pharmafranchisehelp.com, citehr.com
- DRAP (Drug Regulatory Authority of Pakistan) — dra.gov.pk